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Submission link: <https://forms.gle/9QfcWiqewGmCcvG36>

Theory of Computation

Exercise 9: (Context-free grammar part 2)

1. Show that $L(G_1) \neq L(G_2)$.

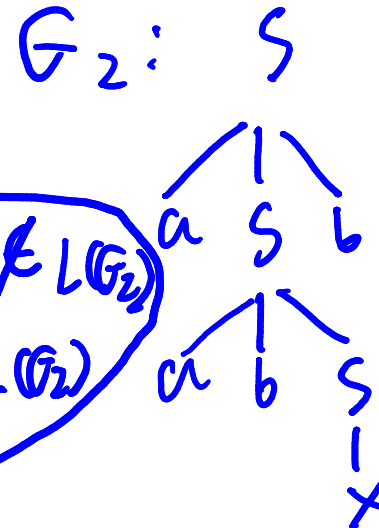
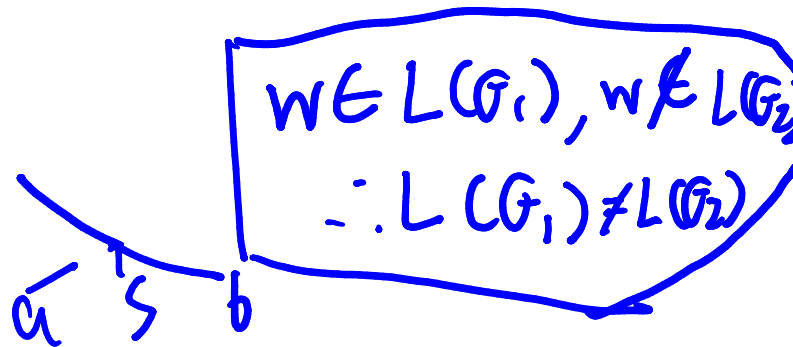
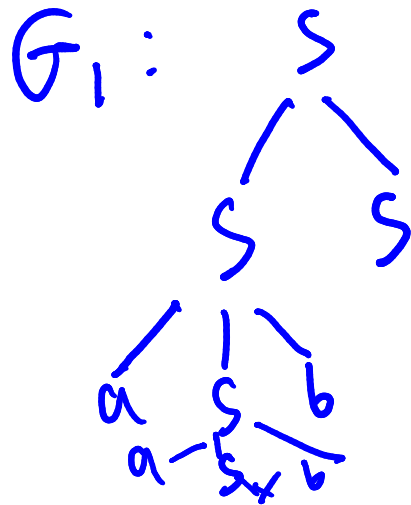
$G_1 = (\{S\}, \{a, b\}, S, P_1)$

$P_1: S \rightarrow aSb \mid SS \mid \lambda$

$G_2 = (\{S\}, \{a, b\}, S, P_2)$

$P_2: S \rightarrow aSb \mid abS \mid \lambda$

$w = aabbaab$



2. Find CFG for the language L.

$$L = \{ a^i b^j c^k : j = i + k \} \Rightarrow \{ a^i b^{i+k} c^k \} \Rightarrow \{ a^i b^i b^k c^k \}$$

$$S \rightarrow XY$$

$$X \rightarrow aXb \mid \lambda$$

$$Y \rightarrow bYc \mid \lambda$$

*3. Use CYK algorithm to find whether **abab** $\in L(G)$.

$G: S \rightarrow AB$

$A \rightarrow BB$

$A \rightarrow a$

$B \rightarrow AB$

$B \rightarrow b$

a	b	a	b
A	B	A	B
ab	ba	ab	
S, B		S, B	
abab	bab		
A	A		
abab			
A			

abab $\notin L(G)$